MA.6.DP.1.1

Benchmark

MA.6.DP.1.1	**Recognize and formulate a statistical question that would generate numerical data.**

Example: The question “How many minutes did you spend on mathematics homework last night?” can be used to generate numerical data in one variable.

Connecting Benchmarks/Horizontal Alignment

- MA.6.AR.1.1

Terms from the K-12 Glossary

- Data
- Statistical Question

Vertical Alignment

Previous Benchmarks

- MA.5.DP.1.1, MA.5.DP.1.2

Next Benchmarks

- MA.7.DP.1.5

Purpose and Instructional Strategies

In previous courses, students collected numerical and categorical data to construct tables and graphs. In grade 6 accelerated, students are formally introduced to statistics and begin formulating statistical questions for numerical data. In future courses, students will extend this when working with categorical data and then with bivariate data *(MTR.1.1, MTR.4.1)*.

- Statistics are numerical data relating to a group of individuals; statistics is also the name for the science of collecting, analyzing and interpreting such data. It is a method for translating a real-world context into a collection of numbers. A statistical question anticipates an answer that varies from one individual to the next and is written to account for the variability in the data. Data are the numbers produced in response to a statistical question. Data are frequently collected from surveys or other sources (i.e., documents).
- Instruction focuses on statistical questions that generate numerical data. Once the data is gathered, it can be organized in a table or displayed in a graph.
- Instruction includes the understanding of a statistical question and a non-statistical question. A statistical question can be answered by collecting information that addresses differences in a population. A statistical question concerns some characteristic of a collection of individuals rather than of one specific individual.

- For example, the question “How tall am I?” is not a statistical question because it is only about a single individual and there is only one response; however, “How tall are the students in my class?” is a statistical question since it can be answered by collecting information from many students and it is anticipated that this information will vary from one student to the next.

Examples of Statistical Questions	**Non-Examples of Statistical Questions**
• What is the range of ages of students in my school? • How many pets are owned by each student in my grade level? • What is the average math test scores of the students in my class? • How many cupcakes of each type were made at the bakery in a week? • How many letters are in the names of each person in my class? • What is the median height of people in my class?	• How old am I? • How many pets do my grandparents own? • What is my sister’s math test score? • What is Martha’s favorite type of cupcake? • How many letters are in my name? • What is my height?

- For numerical data, statistical questions can result in a narrow or wide range of numerical values. To collect information, students design a statistical question that anticipates variability by providing a variety of possible anticipated responses that have numerical answers, such as 3 hours per week, 4 hours per week and so on. Be sure questions have specific numerical answers.
- Statistical questions address the following two aspects: population of interest and measurement of interest. These types of questions should be formulated to anticipate answers that vary.
- Instruction includes students recognizing that statistical questions exist to collect data that can be organized, analyzed and interpreted. This includes determining the measure of center and measure of variation.
  - For example, the statistical question “How much money does a household in Lake County generally spend on the internet in a year?” implies that one must gather numerical data on the cost of internet for households in Lake County to determine how much is generally spent.
- Instruction includes the understanding that statistical questions are not limited to data that can be gathered by students themselves but can also be found by researching or using existing data sets.
  - For example, the question “What was the temperature in Miami in the month of July?” can be considered a statistical question because the data had already been collected.

Common Misconceptions or Errors

- Students may believe that all the questions that they write are statistical questions. Students need to understand that a statistical question must have a range of numerical answers. They will need to have sample questions and practice writing their own.

Strategies to Support Tiered Instruction

- Teacher provides a list of examples of statistical questions and what qualifies them as statistical to assist students who incorrectly label non-statistical questions.
- Teacher provides instruction focused on patterns that make statistical questions and how to change non-statistical questions into statistical questions. Teachers work with students on examples to determine if each question is statistical or non-statistical. If the question is non-statistical, the teacher models how to convert it to a statistical question.
- Teacher provides a list and co-creates a graphic organizer that determines the pattern from statistical and non-statistical questions. Students will have an opportunity to explain why each one is either statistical or non-statistical.
  - Example of a graphic organizer is shown below.

Question Type	Characteristics	Examples
Statistical	A question that can be answered by collecting data. Often there will be variability in the data.	What is the average cost of a home in this neighborhood?
Non-statistical	A question that is typically not answered by collecting data or used to collect data.	When will you be home?
Data-gathering but non-statistical	Often a question that is on a survey.	Does your father work from home?

- Teacher provides sample questions and models how to practice writing their own.

Instructional Tasks

Instructional Task 1 (MTR.7.1)

Write a statistical question that will require data that could be collected from your class to provide information to an advertising agency for teenagers to purchase clothing, shoes, backpacks or school supplies.

a. Design a table for data collection.
b. Create a proposal to the advertising agency to recommend an item to advertise based on the data you collected from your statistical question. Include data to support your recommendation.

Instructional Items

Instructional Item 1

Amir collected data from his sixth-grade class at Liberty Middle School. He asked about the amount of time students spent playing video games. Write a statistical question associated with his data.

Instructional Item 2
Which of the questions below are statistical?

a. Are there typically more manatees in Tarpon Springs or in Blue Springs?
b. How many Instagram followers does Mr. Youmans have?
c. How many points did the basketball team score in its last game?
d. What is the range of photos students in Ms. Alvarez's class have on their phone?
e. How many minutes a week do 8^{th} graders at Bridgton Middle School spend watching TV?

**The strategies, tasks and items included in the B1G-M are examples and should not be considered comprehensive.*